

Student Performance Predictor

Predictive analytics interface for academic trajectories and intervention resource allocation.

Prediction Interface **System Metrics & KPIs**

Model Architecture & KPIs

This dashboard is powered by a **Gradient Boosting Regressor**, chosen for its robustness to outliers and high variance capture.

Mean Absolute Error (MAE)

0.69

↓ - Target: < 3.0

R-Squared (R^2)

0.74

↑ + Target: > 0.70

Inference Latency

< 100ms

↓ - Target: < 500ms

Business Value Mapping:

- Early Intervention (Business KPI):** Identifies students predicting below 65, allowing targeted resource allocation to prevent academic failure.
- Data Completeness (Data Quality KPI):** 100% of missing values seamlessly handled via Streamlit-to-Scikit-Learn median/mode imputation pipeline.